ARCHITECTURE DECISION RECORD · A NIGERIAN PAYMENT PROCESSOR

4 accepted · 1 proposed

Payments gateway: hybrid modernisation

Decision requested: Endorse ADR-0001 to ADR-0004, and ask the compliance team to decide ADR-0005 before the foundations phase starts.

05 decisions recorded

Version 1.0 · 18 October 2026 · Sakariyau Oseni

PUBLIC · PORTFOLIO CASE STUDY

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A

Executive summary

Move three products to AWS, keep the record in Lagos, and make sure every payment is delivered once and never lost.

Problem

One on-premises table on every payment's path, a duplicate check that reads and then acts, and no rule for a vendor timeout.

Recommendation

Claim each payment once in DynamoDB, stream every change to Lagos through Amazon MSK, re-query unknown outcomes, keep state revenue on premises, run in Cape Town (proposed).

3M

payments a day; 300 a second at peak

0

payments delivered twice, by design

$2,690

a month on demand; $0.03 per 1,000 payments

27/27

tests passed locally, in CI and on AWS

DECISION REQUESTED Endorse ADR-0001 to ADR-0004, and ask the compliance team to decide ADR-0005 before the foundations phase starts.

B

Five pressures, eight requirements

01 One transaction table on every payment's path

02 Duplicates from retries: USSD drops, terminal timeouts, double taps

03 Vendor timeouts leave the outcome unknown

04 One data centre, capacity bought for the peak

05 No AWS Region in Nigeria; payment records stay in Lagos

ID	MEASURABLE REQUIREMENT
REQ-01	Volume: 3 million payments a day; 300 a second at peak
REQ-02	Duplicates: none delivered, ever
REQ-03	Durability: no accepted payment lost (recovery point zero)
REQ-04	Latency: gateway adds at most 300 ms at p95, vendor time excluded
REQ-05	Availability: 99.95% a month for the cloud tier
REQ-06	Record: 99% of changes in Lagos within 5 s; none lost in a 7-day link outage
REQ-07	Unknowns: settled within 15 minutes, or passed to operations
REQ-08	Residency: claims held 48 hours, events 7 days, nothing longer

B

Scope and assumptions

In scope

- Electricity, airtime and data, and pay-TV moved to AWS
- State revenue routed to Lagos, unchanged
- Syncing every change to the system of record
- Network, security, cost and migration order

Out of scope

- Channels' own systems and card acquiring
- Settlement and reconciliation engines
- A second AWS Region

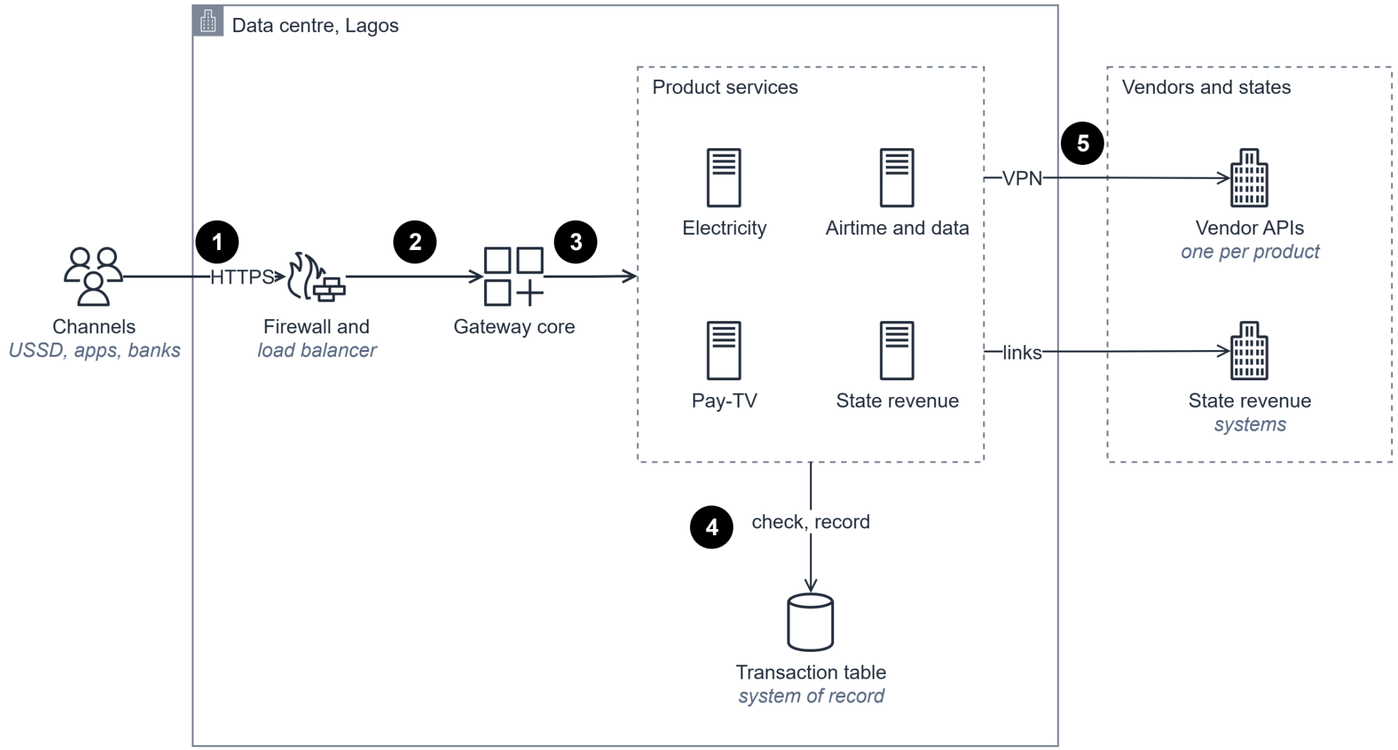
Assumptions

- Every vendor offers a status query and logs every request it accepts (to go into contracts)
- State integrations trust the data centre's addresses (to be confirmed by an inventory)
- Volumes and latency targets are mine: the original analysis left them blank
- Compliance decides where payment data may live; ADR-0005 waits on it

C

Current state: one table on every payment's path

FIGURE 1 · CURRENT STATE, ONE DATA CENTRE IN LAGOS



1 Channels reach the data centre over HTTPS

2 A firewall and load balancer pass requests to the gateway core

3 The core routes each payment to its product service

4 Every product checks the one table for duplicates, then records state there: two trips per payment

5 Product services call vendors over VPN; state revenue uses its own links

C

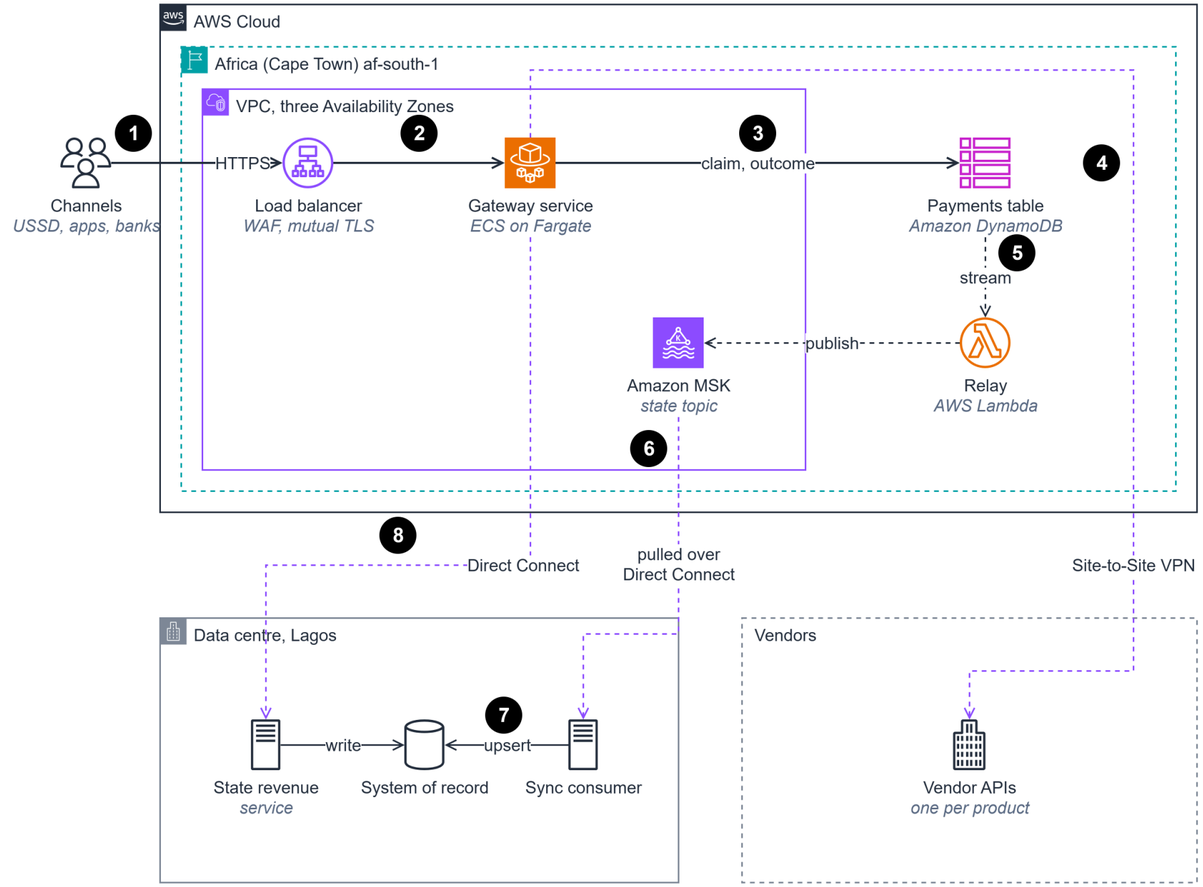
Current-state findings

ID	AREA	FINDING	SEVERITY	ADDRESSED BY
F1	Correctness	Duplicate check reads, then forwards in a separate step	**Critical**	ADR-0001
F2	Correctness	No rule for a vendor timeout	**High**	ADR-0004
F3	Scalability	One table on every payment's path, twice	**High**	ADR-0001, ADR-0002
F4	Resilience	One data centre, capacity bought for the peak	**High**	Target state
F5	Operations	Throughput and latency never measured	**Medium**	REQ-01, REQ-04

D

Target state: three products in AWS, the record in Lagos

FIGURE 2 · TARGET STATE, CAPE TOWN AND LAGOS



1 Channels send HTTPS with a client certificate; WAF and the load balancer check it

2 The gateway service on ECS with Fargate validates and routes

3 One conditional write claims the payment; its outcome is recorded on the same item

4 The vendor is called over its own Site-to-Site VPN

5 DynamoDB Streams and the Lambda relay publish each change to Amazon MSK

6 The sync consumer in Lagos pulls the topic over Direct Connect

7 It writes the system of record, newer versions only

8 State revenue goes to Lagos over Direct Connect, as today

D A payment’s states and failure handling

PENDING	Claimed; the vendor is being called
SUCCESS	Delivered; the token is stored for duplicates
FAILED	Declined, or no vendor record when the 15-minute window closes
UNKNOWN	Timed out after sending: re-queried after 30 s, 1, 2, 4 and 8 min
REVIEW	The vendor keeps erroring: operations and reconciliation settle it

FAILURE	WHAT HAPPENS
20 identical requests at once	One claim wins; all 20 get the same answer
The vendor answers too late	UNKNOWN, then SUCCESS from the re-query; one delivery
A task stops mid-payment	The stale PENDING claim is re-queried after 60 s
Direct Connect fails	Backup VPN; the three cloud products continue
The consumer is down	Changes wait on the topic up to 7 days, then catch up

E

Decision log

ID	DECISION	STATUS	REQUIREMENTS
ADR-0001	Claim each payment once with a DynamoDB conditional write	**Accepted**	REQ-02, REQ-03, REQ-04
ADR-0002	Sync the system of record through DynamoDB Streams and Amazon MSK	**Accepted**	REQ-03, REQ-06
ADR-0003	Keep state revenue payments on premises	**Accepted**	REQ-06
ADR-0004	Resolve unknown outcomes by re-querying, never by re-sending	**Accepted**	REQ-02, REQ-07
ADR-0005	Run the cloud tier in Cape Town, holding payment data for at most seven days	**Proposed**	REQ-05, REQ-08

E

ADR-0001 · Claim each payment once with a DynamoDB conditional write

CONTEXT AND OPTIONS

Channels retry and sometimes send the same payment twice at once. Today's check reads the table, then forwards in a separate step, so two identical requests can both be sent.

Options: DynamoDB conditional write (chosen); Redis SET NX in ElastiCache; Read the cache, then forward; Synchronous check in Lagos

DECISION

Accepted

We will claim each payment with one write that succeeds only if the reference is new or its 48-hour claim has expired.

Confidence High
Requirements REQ-02, REQ-03, REQ-04

CONSEQUENCES · GOOD

- Atomic: of any number of duplicates, one claim wins
- Durable once acknowledged, across Availability Zones
- Duplicates get the original answer back

CONSEQUENCES · BAD

- Single-digit milliseconds, not sub-millisecond
- Payment data held in the cloud for the window (ADR-0005)

CONFIRMATION

20 identical requests at once produce one delivery (demo and AWS run)

E

ADR-0002 · Sync the system of record through DynamoDB Streams and Amazon MSK

CONTEXT AND OPTIONS

The system of record stays in Lagos. Services that write the state and then publish it lose the change if they stop between the two writes.

Options: Streams, Lambda relay, MSK (chosen); Each service publishes to Kafka; Kinesis Data Streams for DynamoDB; Synchronous writes to Lagos

DECISION

Accepted

We will write each change once to DynamoDB; the stream and a Lambda relay publish it to MSK, keyed by payment, and Lagos pulls it.

Confidence Medium
Requirements REQ-03, REQ-06

CONSEQUENCES · GOOD

- No dual write; order kept per payment
- Payments continue through a 7-day link outage
- The record's zone opens no inbound port

CONSEQUENCES · BAD

- MSK is the largest line: about $630 a month
- The record lags by seconds; 24-hour stream window to watch

CONFIRMATION

Consumer stopped: 5 payments arrive after restart; a second replay changes nothing

E

ADR-0003 · Keep state revenue payments on premises

CONTEXT AND OPTIONS

State revenue integrations are built one state at a time, over links that trust the data centre's addresses.

Options: Keep them in Lagos, routed over Direct Connect (chosen); Move them with the other products; Split traffic before the gateway

DECISION

Accepted

We will leave the integrations and their writes in Lagos; the cloud gateway routes state revenue there.

Confidence Medium
Requirements REQ-06

CONSEQUENCES · GOOD

- No state has to change its side
- Revenue records are in the system of record at once

CONSEQUENCES · BAD

- Two duplicate checks
- Stops if Direct Connect and the backup VPN both fail

CONFIRMATION

An inventory of each state's dependencies; a failover test over the backup VPN

E

ADR-0004 · Resolve unknown outcomes by re-querying, never by re-sending

CONTEXT AND OPTIONS

A timeout leaves the gateway not knowing whether the vendor delivered. Re-sending can pay twice; failing can refund a delivered token.

Options: Re-query until the window closes (chosen); Re-send with the same reference; Fail at once and reverse; Leave it to end-of-day reconciliation

DECISION

Accepted

We will record UNKNOWN, answer PENDING, and re-query for 15 minutes; never send the same payment twice.

Confidence High
Requirements REQ-02, REQ-07

CONSEQUENCES · GOOD

- No payment delivered twice
- Most unknowns settle within minutes

CONSEQUENCES · BAD

- Channels must handle PENDING
- Relies on vendors logging every request (contracts)

CONFIRMATION

A late vendor ends SUCCESS with one delivery; a lost request ends FAILED with none

E

ADR-0005 · Run the cloud tier in Cape Town, holding payment data for at most seven days

CONTEXT AND OPTIONS

No AWS Region in Nigeria; the Lagos Local Zone has no DynamoDB, MSK or Fargate. Where payment data may live is for compliance to decide.

Options: af-south-1, 7-day retention (proposed); Lagos Local Zone, self-managed stores; A European Region; Stay on premises

DECISION

Proposed

We propose af-south-1, with claims expiring after 48 hours, events after 7 days, and the record kept in Lagos.

Confidence Medium
Requirements REQ-05, REQ-08

CONSEQUENCES · GOOD

- Managed services in three Availability Zones
- No payment record in the cloud beyond 7 days

CONSEQUENCES · BAD

- At least 96 ms of each payment is the distance
- Depends on compliance's written approval

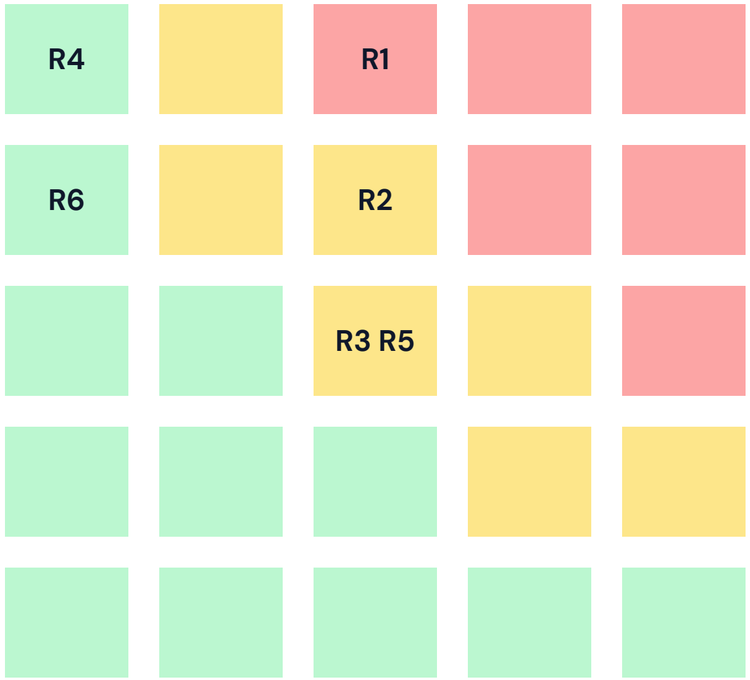
CONFIRMATION

Written approval before production; latency test from Lagos; SCP and Config rule

F

Risks and mitigations

LIKELIHOOD 1 TO 5, LEFT TO RIGHT
IMPACT 1 TO 5, BOTTOM TO TOP



ID	RISK AND MITIGATION	OWNER AND SCORE
R1	Compliance refuses data in Cape Town Lagos Local Zone fallback, self-managed stores	Compliance 15 · High
R2	A vendor cannot answer status queries Contract clause before cut-over; REVIEW path	Vendor management 12 · Moderate
R3	Added latency over 300 ms at p95 Load test from Lagos before each product moves	Architecture 9 · Moderate
R4	Relay broken for over 24 hours Alarm on iterator age at 5 min; daily reconciliation	Operations 5 · Low
R5	Team new to Kafka and DynamoDB Training; partner support for the first quarter	Engineering 9 · Moderate
R6	Direct Connect and backup VPN both fail Two Direct Connect ports; cloud products continue	Network 4 · Low

F

Monthly cost estimate, indicative

LINE	USD / MONTH
Amazon MSK · 3 kafka.m7g.large brokers, 300 GB	$631
AWS Direct Connect · 2 dedicated 1 Gbps ports in Lagos	$460
AWS Transit Gateway · 8 attachments, 600 GB processed	$362
AWS Site-to-Site VPN · 6 vendors and 1 backup to Lagos	$256
Amazon ECS on AWS Fargate · 6.5 Graviton vCPUs, 13 GB	$253
Route 53 Resolver endpoint · 2 network interfaces	$183
Interface VPC endpoints · 6 services in 3 zones	$173
Amazon DynamoDB · 180 million writes, PITR	$158
WAF, ALB, transfer, logs, KMS, Lambda	$218
TOTAL	**$2,690**

$2,690

a month on demand in af-south-1

Assumptions

- 3 million payments a day: $0.03 per 1,000
- On-demand prices, AWS Price List API, 18 October 2026
- Connectivity is 47% of the bill
- Excludes support, colocation and vendors' VPN ends

G

Proof: demo and live run on AWS

27 of 27

tests passed against DynamoDB, the Lambda relay and Amazon MSK on AWS

75 / 0

relay invocations / errors in the live run

2.1 s

longest a change waited in the stream before it was published

$0.35

for the two-hour run, then all 42 resources destroyed

What testing caught

01 A token-provider import that would have crashed on MSK

02 A replay test that assumed a fresh topic: replay restored 13 missed payments

03 A consumer that held its Kafka partitions for 45 s after SIGTERM

G

Roadmap: one product at a time

PHASE 0 Foundations	PHASE 1 Pay-TV	PHASE 2 Airtime and data	PHASE 3 Electricity	PHASE 4 Retire fallback
Compliance decision; landing zone; Direct Connect, VPNs, Transit Gateway Exit: ADR-0005 decided; links tested	Pay-TV through the cloud gateway Exit: A month end with REQ-01 to REQ-07 met	Exit: Load test at 300 a second	Exit: Vendor status queries confirmed	Retire the on-premises fallback Exit: All three products through a month end

H Next steps

DECISION REQUESTED

Endorse ADR-0001 to ADR-0004, and ask the compliance team to decide ADR-0005 before the foundations phase starts.

01 Compliance decides ADR-0005 and the data classification

02 Vendor management adds status-query and request-logging clauses

03 Network orders two Direct Connect ports in Lagos

04 Engineering starts Kafka and DynamoDB training

Thank you

Every payment delivered once, and never lost.

Sakariyau Oseni

github.com/ooseni/TESA-Cloud-Engineering